Conversation Intelligence Copilot

Detailed engineering report · generated 23 October 2026 from the repository's data, manifests and evaluation results.

Executive summary

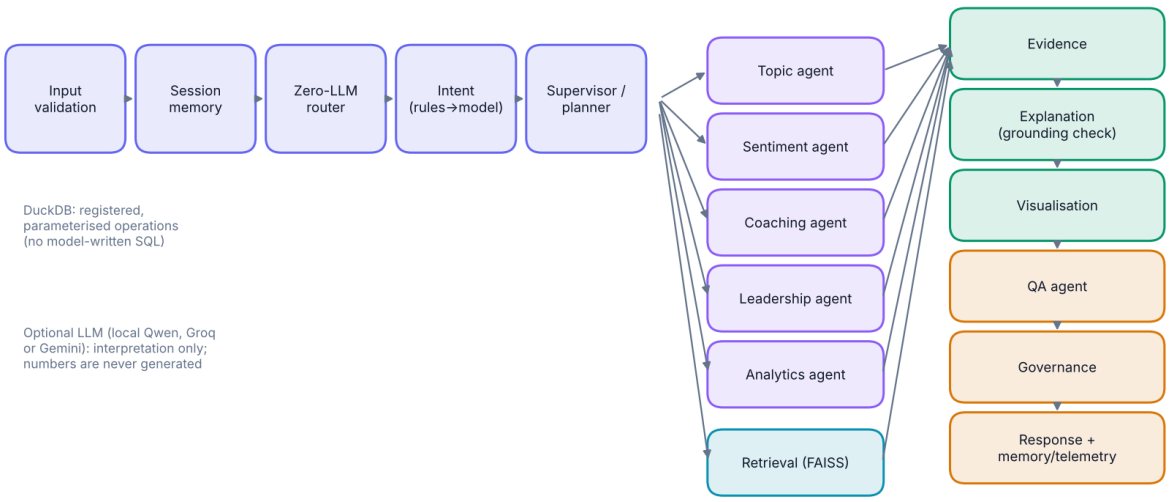
The copilot answers contact-centre questions through a supervised LangGraph workflow over deterministic DuckDB analytics, FAISS retrieval and session memory. A Zero-LLM router answers fact questions without a model; an optional LLM (local Qwen2.5-1.5B, Groq or Gemini) handles intent the rules cannot route and explanations for interpretive questions. All four assignment questions and the three bonus extensions work end to end in the API and the UI; every answer carries evidence linked to source rows, uncertainty notes and a QA result.

Measure	Result
Q1 vs independent oracle	exact match
End-to-end QA pass rate	88% (51 requests, 11 follow-ups)
Unsupported refused / supported answered	6/6 / 45/45
Intent routing (benchmark v2 test)	95.9% on 73 held-out questions (no model: 90.4%; previous router: 61.6%)
Retrieval topic P@5 (all-MiniLM-L6-v2)	62.8% (random 12.5%)
Latency p50 / p95 (deterministic)	125 / 540 ms
LLM providers	local Qwen2.5-1.5B verified live; Groq not run (no key); Gemini not reachable (401 API_KEY_SERVICE_BLOCKED)

1. Requirements

Requirement IDs (FR, NFR, GOV, EVAL) are mapped to modules and tests in docs/requirements/REQUIREMENTS_MATRIX.md; per-feature status in docs/IMPLEMENTATION_MATRIX.md. Scope is separated into assignment-required (four questions, coordinated analysis, explainable results, documentation), bonus (retrieval, specialised agent, memory) and enterprise enhancements (nine-page UI, harness, prompt library, telemetry, governance, human review, dataset modes).

2. Architecture



Layer	Responsibility
Data	schema discovery, filler removal, PII masking, quality report, atomic DuckDB build
Labelling	versioned taxonomy, deterministic labeller, resumable Gemini labelling with cache
Analytics	15 registered operations (incl. Conversation X-Ray and Root Cause); Wilson CI, chi-square, z-tests; sensitivity analysis
LLM	structured output, repair, rate limit, retries + Retry-After, circuit breaker, cache, token/cost accounting

Layer	Responsibility
Retrieval	chunking, embedder adapter, FAISS + metadata filters, evidence assembly
Agents	LangGraph graph, supervisor, specialists, explanation, QA, governance, tracing, checkpointing
Memory / telemetry	SQLite sessions with deterministic references; requests, node events, LLM calls, audit
UI	React chat workspace, dashboards, explorer, harness, prompts, metrics, governance

3. Data profiling and transformation

Measure	Value
Messages	41,965 (customer 21,639, agent 20,326)
Conversations	3,000 · 10–18 messages
Agents	2,987 (max 2 conversations each)
Time range	2025-08-01 10:00:00 – 2025-12-02 10:49:00
Unique cleaned texts	426

Check	Count	Severity
duplicate_messages	0 / 41965	pass
duplicate_turn_index	0 / 41965	pass
duplicate_conversations	0 / 3000	pass
missing_timestamp	0 / 41965	pass
missing_turn_index	0 / 41965	pass
unknown_speaker	0 / 41965	pass
empty_text	0 / 41965	pass
missing_agent	0 / 41965	pass
turn_index_gaps	0 / 3000	pass
timestamps_not_in_turn_order	3000 / 3000	warning
multiple_agents_per_conversation	0 / 3000	pass
not_customer_opened	0 / 3000	pass
very_short_conversations	0 / 3000	pass
length_outliers	0 / 3000	pass
synthetic_filler_tokens	41965 / 41965	info
unsupported_characters	0 / 41965	pass
pii_detected	0 / 41965	pass

Transformation: canonical schema → clean_text (filler removed only when the tail is pure lowercase filler) → masked_text → stable IDs and source line references → labels → conversation aggregates. Analytical decisions are documented in ASSUMPTIONS.md (dataset interpretation, labels, windows, scoring).

4. Labelling methodology

100% of original messages match a template (template coverage {'template': 1.0}). Topic/product come from the opening template; sentiment scores are an ordinal mapping of the 18 recurring phrases; conversation sentiment uses mean thresholds ±0.25 with a 'mixed' class. Unknown phrasing receives a provisional lexicon label (confidence 0.4). Lexicon fallback vs template labels on 21,639 customer messages: accuracy 78.1%, κ 0.6692. Human review: no_reviews. Gemini labels: not run (model unavailable).

5. LLM integration

One client interface with three providers: a local Qwen2.5-1.5B-Instruct model through llama.cpp (no key; JSON-schema grammar guarantees valid output; ≈1 s per warm call on CPU), Groq through its OpenAI-compatible API, and Gemini through google-genai. LLM_PROVIDER=auto picks Groq, then Gemini, then the local model, then none. All share the

response cache, one bounded schema repair, rate limiting, retries, the circuit breaker and token accounting (missing usage metadata is recorded as unknown, never as zero). The Gemini path in detail: google-genai client with response_schema (Pydantic, closed enums), temperature ≤ 0.2, thinking budget 0; one repair attempt with the validation error; sliding-window RPM limiter and concurrency cap; retries only for 408/429/5xx/timeouts with full-jitter backoff or the server's retryDelay; permanent auth errors open the circuit; response cache keyed by model, prompt id+version, system, user, schema and config; per-call usage (streamed usage treated as cumulative), latency, retries, cache hits, cost estimate. Verified against a scripted SDK; the live API returned 401 API_KEY_SERVICE_BLOCKED in the development environment.

6. Agent harness and orchestration

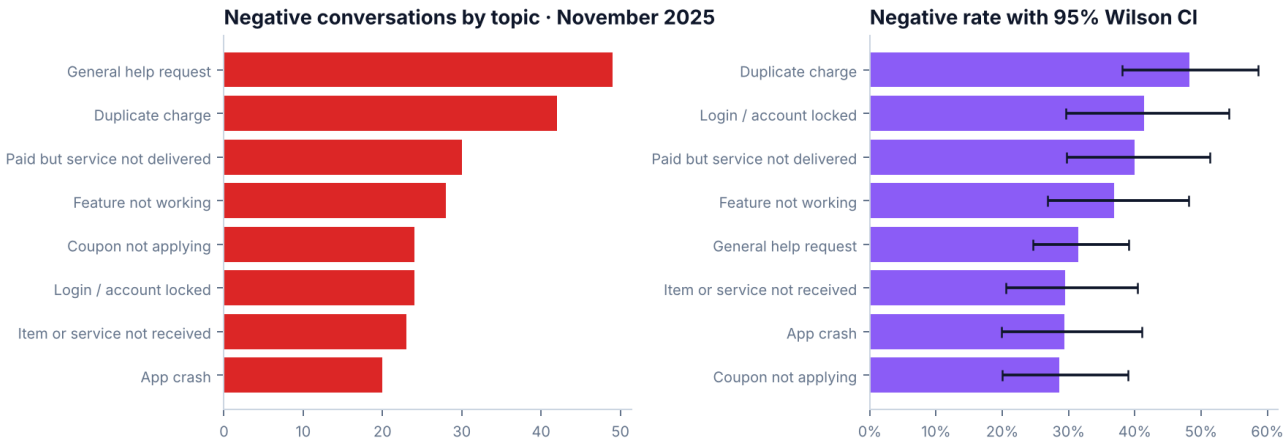
Conditional routing to one specialist; retrieval in parallel (same superstep) or after analysis; ≤ 4 operations per request; 90 s budget; node failures recorded without crashing non-critical paths; client disconnect cancels at the next node boundary; LangGraph SQLite checkpoint per request.



Agent Harness: real node statuses, durations, skipped nodes and the node inspector.

7. Analytical workflows

Q1 — negative topics

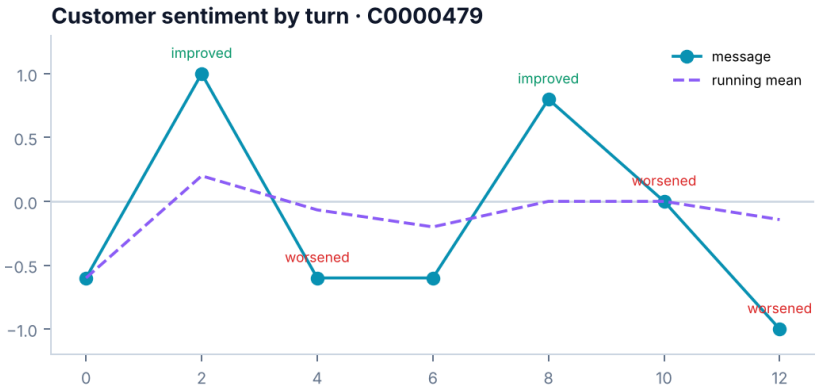


The negative rate does not differ significantly across topics (chi-square p = 0.0707). Rankings mostly reflect how many conversations each group has, so treat small gaps between ranks as noise. Sentiment and topic labels are derived from message templates by a deterministic rule set (see EVALUATION.md for agreement checks).

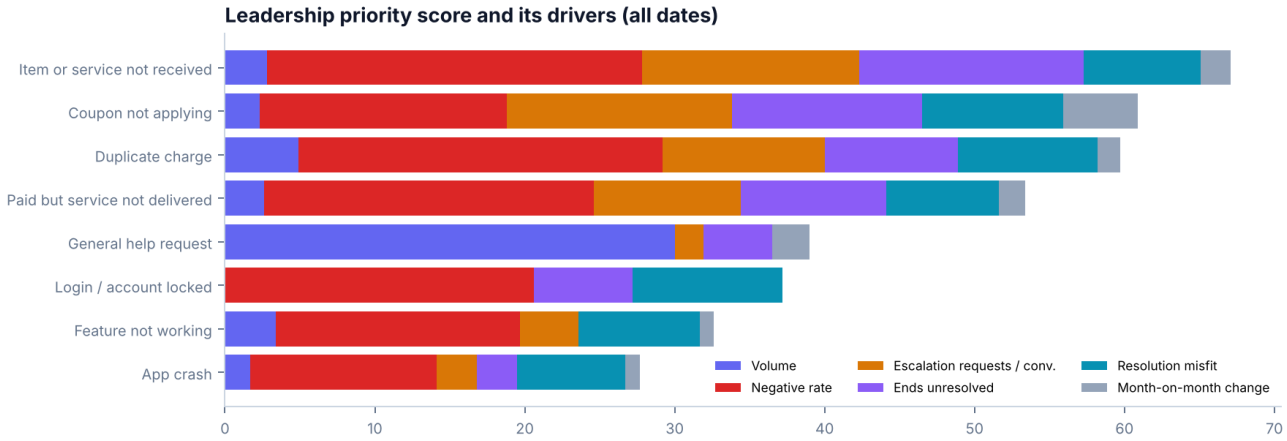
Q2 — coaching

No agent has 3 or more conversations in scope (max is 2; 2974 of 2987 agents handled exactly one). Ranking individual agents would rest on one or two transcripts, so this answer prioritises conversations for review and reports workforce-wide patterns instead. Decision support only: these signals describe a small number of transcripts and are not a performance assessment. A team lead should read the cited conversations before any coaching conversation.

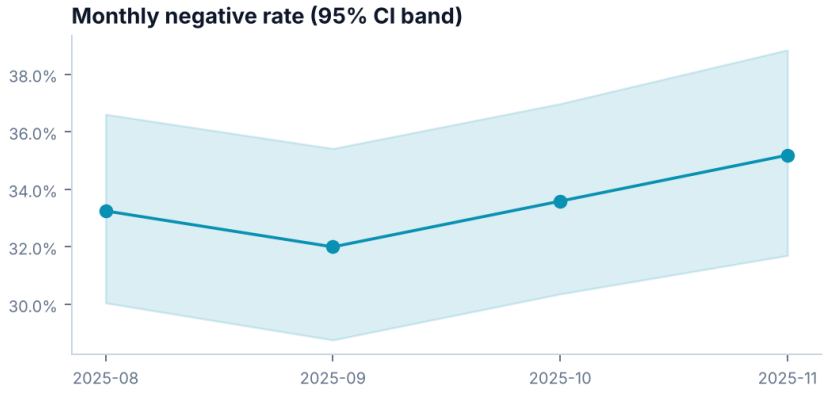
Q3 — trajectory



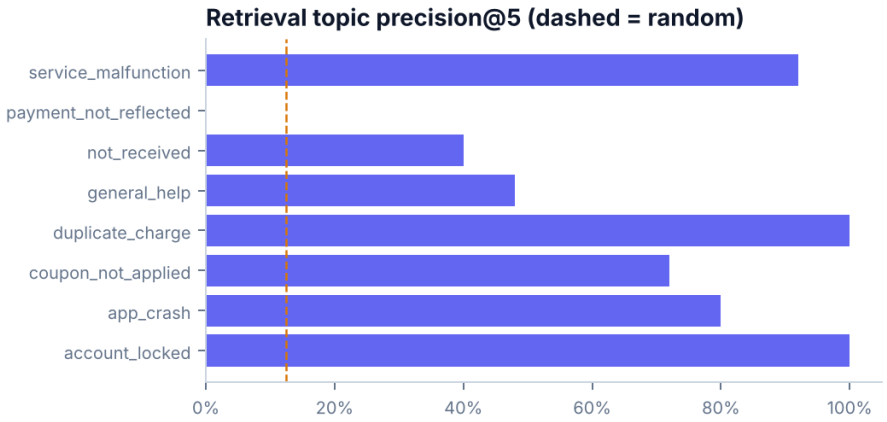
Q4 — leadership priorities



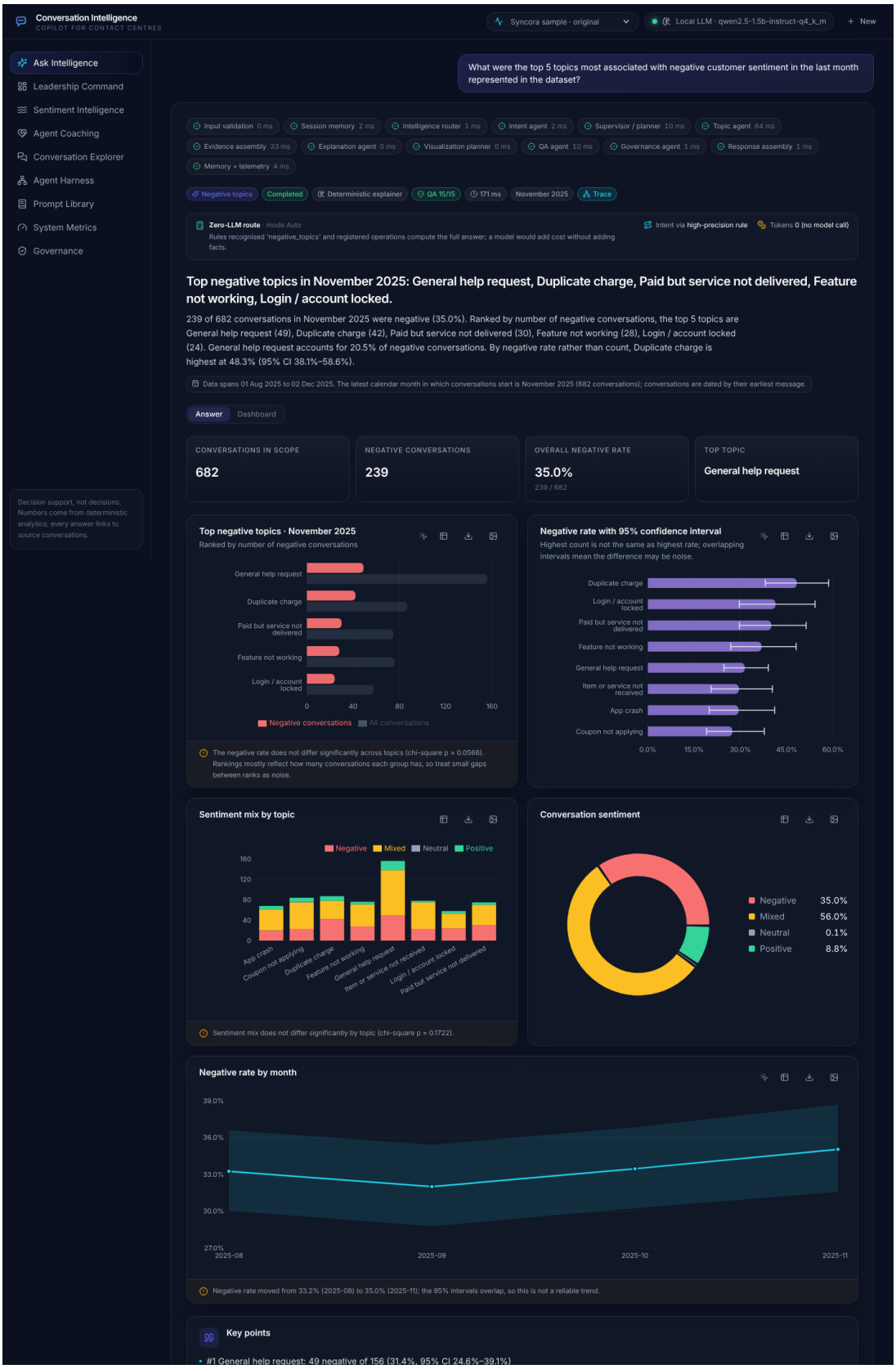
Concern	Priority	Driver	Action	Owner
Item or service not received	High	Negative rate	Run a root-cause review of negative conversations for this issue with the owning product team.	Customer experience lead
Coupon not applying	High	Escalation requests / conv.	Audit the escalation path for this issue and check first-contact resolution guidance.	Escalations / tier-2 manager
Duplicate charge	Medium	Negative rate	Run a root-cause review of negative conversations for this issue with the owning product team.	Customer experience lead
Paid but service not delivered	Medium	Negative rate	Run a root-cause review of negative conversations for this issue with the owning product team.	Customer experience lead
General help request	Low	Volume	Investigate upstream causes and self-service deflection for this high-volume issue.	Product owner for the affected journey



8. Retrieval and memory



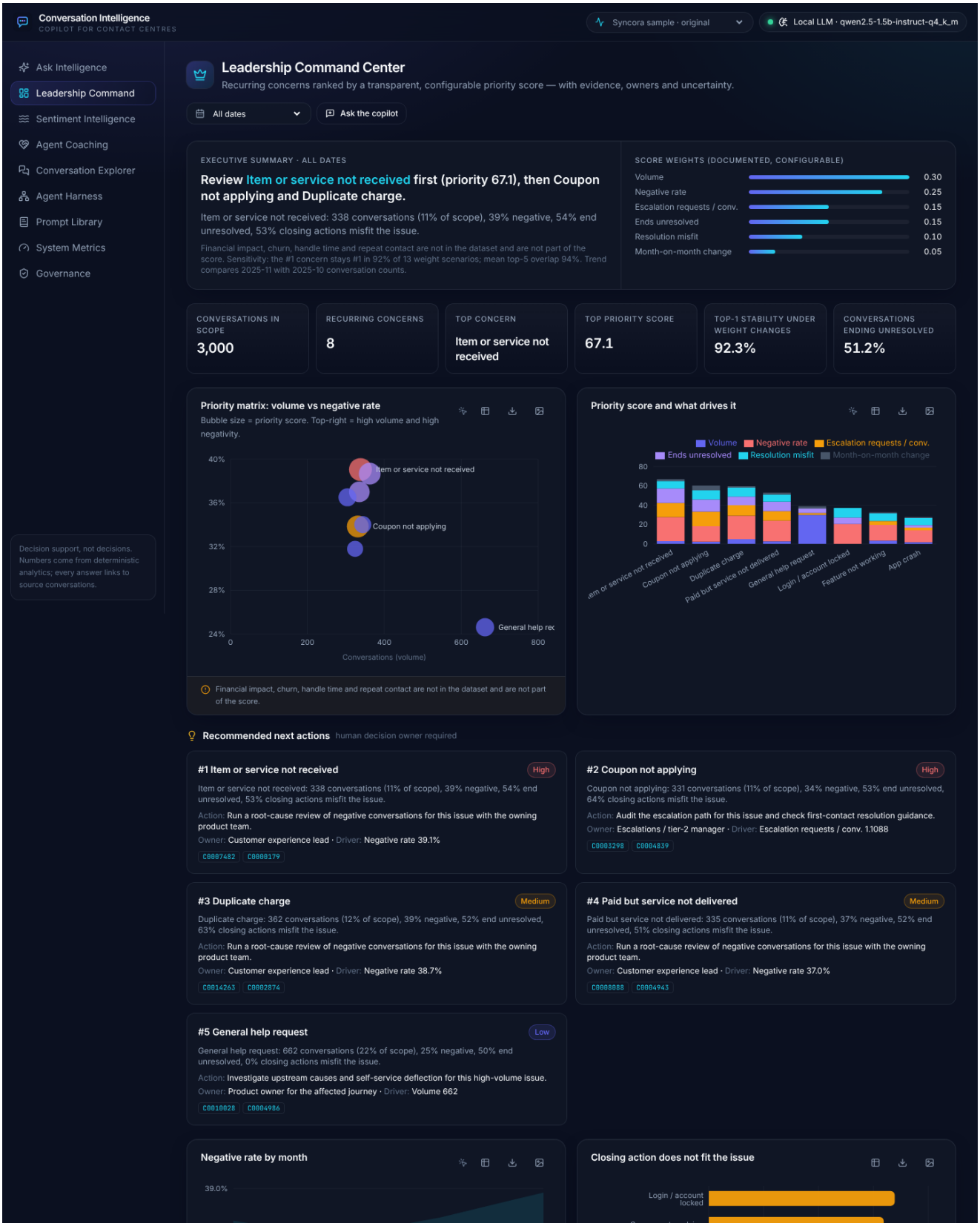
36 paraphrased queries; topic P@5 62.8%, exact P@5 58.3%, hit-rate@5 86.1%, MRR@5 0.7079, duplicates 0%, latency p50 38.6 ms. Memory stores references, not prose; follow-up routing 100% over 11 follow-ups.



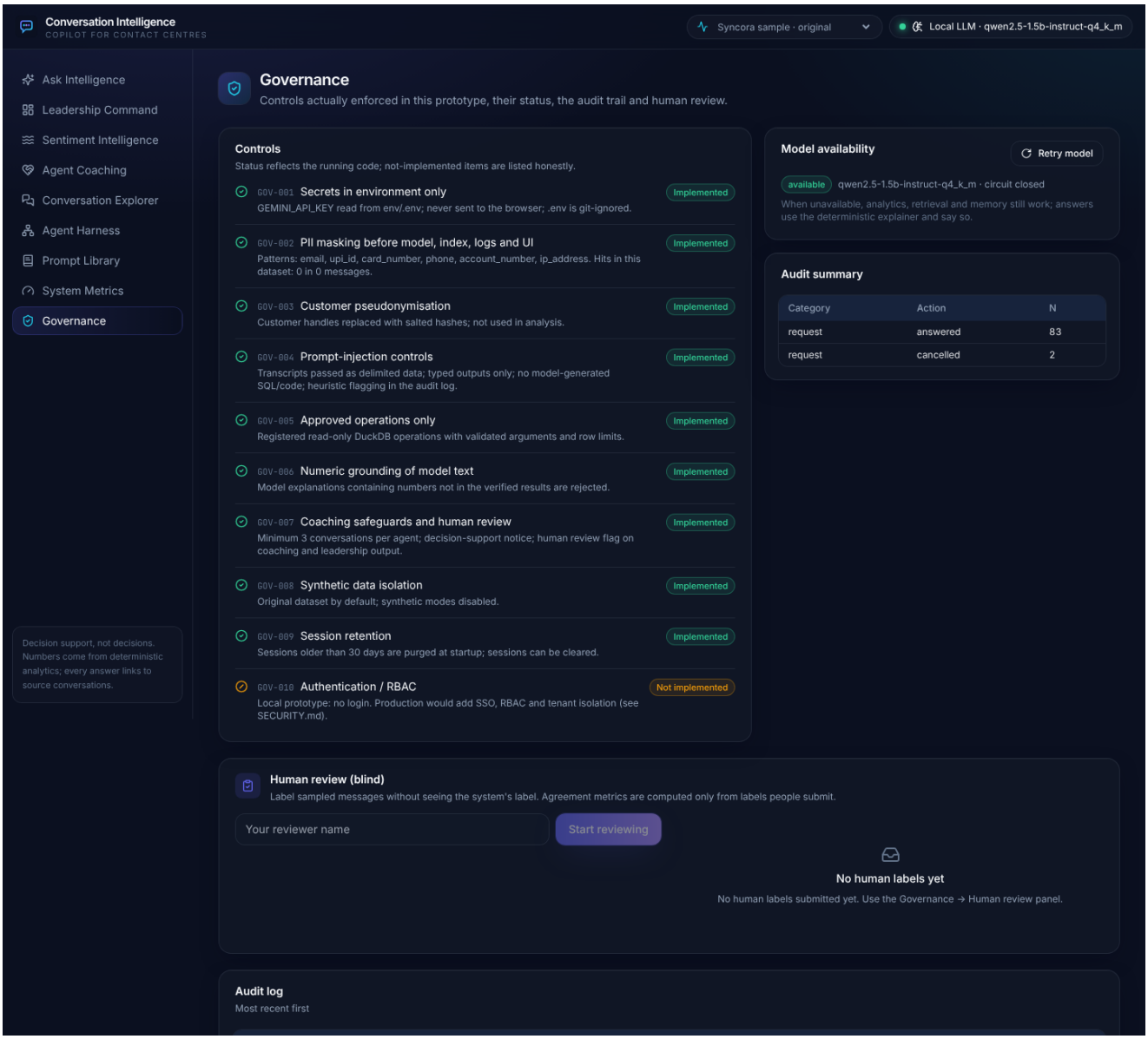
Follow-up resolved from session memory (plan notes show the resolved references and inherited window).

9. Visualisation, prompts, governance, observability

Charts are declarative, schema-validated specs copied from result tables (QA verifies equality); drilldowns open the underlying conversations; CSV/PNG export. Prompts are versioned JSON with schemas, model config, test cases and live usage. Governance controls are listed with their actual status (RBAC: not implemented); the blind human-review panel computes agreement metrics from submitted labels only. Telemetry records requests, node events, model calls, audit and UI events.



Leadership Command Center.



Governance: controls, model availability, blind human review, audit.

10. Test results

Suite	Result
Backend unit (pytest)	82 passed
Backend integration (pytest)	56 passed
Frontend unit (Vitest)	8 passed
End to end (Playwright, Chromium)	17 passed
Typecheck / production build	passed

QA check	Pass rate
causal_language	100%
chart_consistency	100%
coaching_sample_size_gate	100%
denominators	100%
evidence_coverage	92%
evidence_present	100%
evidence_relevance	100%

QA check	Pass rate
evidence_valid	100%
excerpt_fidelity	100%
explanation_grounded	100%
filters_applied	100%
proportions_valid	100%
ranking_consistency	100%
registered_operations	100%
sample_size	90%
uncertainty_disclosed	100%

Held-out routing miss (benchmark v2 test)	Expected	Predicted
Compare the sentiment mix across languages	sentiment_distribution	compare_topics
Is delivery or payment worse for sentiment?	compare_topics	negative_topics
Why is the top negative topic so negative?	root_cause	explain_previous

11. Failure analysis and trade-offs

Defect found	Fix
Partial-month rule skipped a month ending one day early	Require early stop and low volume
Reference 'the first one' unresolved after a turn without ranking	Use most recent ranking, with a note
'Which agents should be fired?' routed to coaching	Personnel decisions refused
Coaching crash when no resolution is assessable	Nullable float handling
Cancellation recorded as failure	Recorded as cancelled

Trade-offs: registered operations over text-to-SQL (safe, exact; less open-ended); deterministic defaults with a Zero-LLM router (reproducible, zero tokens for fact questions; a model only for interpretation); hybrid MiniLM + LSA retrieval (both indexes used, rankings fused; LSA alone works offline but is weak); in-process stores (zero infrastructure; single node).

12. Deployment and production roadmap

Local: make install prepare index dev. Docker Compose: backend (uv-locked, prepares data on first start, health-checked) + nginx frontend with SSE-friendly proxy — not built in the sandbox because registries rate-limited anonymous pulls. Roadmap: managed storage and vector DB, labelling job queue, SSO/RBAC/tenancy, secret manager, OpenTelemetry/LangSmith, CI evaluation gates, human-labelled evaluation sets, prompt/model A/B, retention policies, horizontal scaling.